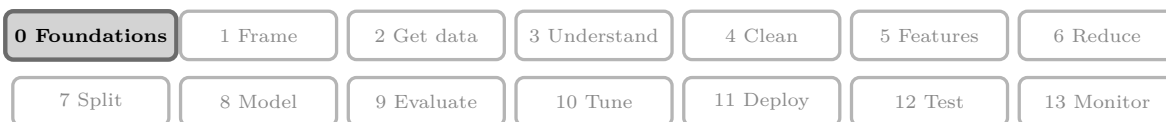


The Hessian and Multivariate Taylor Series

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Eigenvectors and eigenvalues ([Note MA-056](#)); Derivatives of one variable ([Note MA-061](#)); Partial derivatives and gradients ([Note MA-062](#)).
- **Leads to:** Convex and non-convex loss ([Note DL-017](#)).

1. Overview

Key point: The Hessian collects all second partial derivatives and measures how a surface curves; together with the gradient it builds the multivariate Taylor polynomial, which approximates a function near a point by a plane, then a curved bowl, and so on.

This Note follows Chapter 5 (Sections 5.7 and 5.8) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

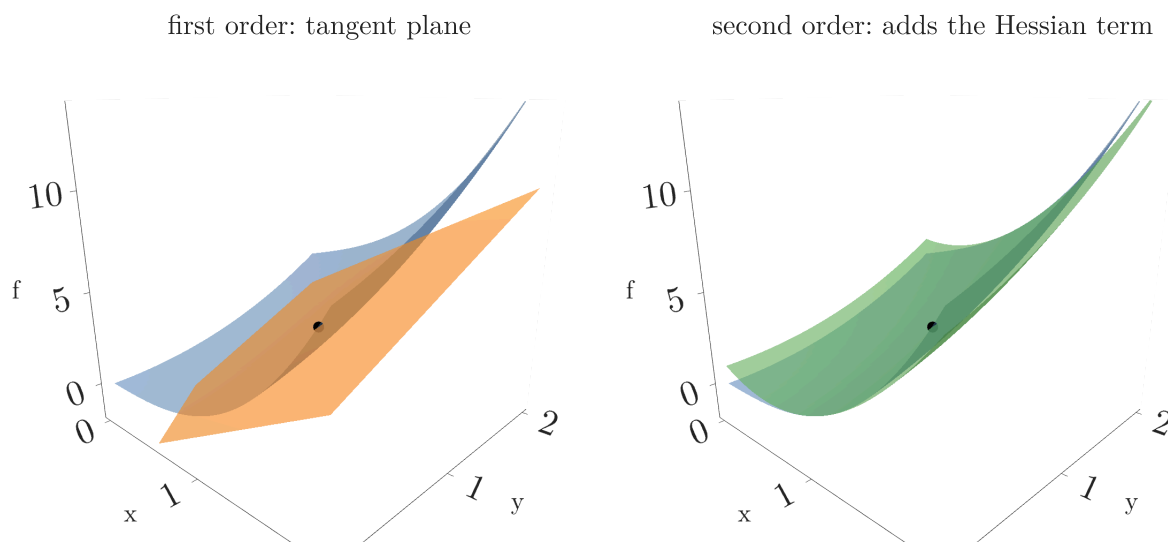


Figure 1: The function $f(x, y) = x^3 + xy + y^2$ (blue) near the point $(1, 1)$, approximated by its tangent plane (left, orange) and by its second-order Taylor polynomial (right, green)

Figure 1 shows the goal. Near a point, a surface is approximated first by a flat **tangent plane** (G-1946; left), which only uses the gradient. Adding a term built from second derivatives, the Hessian, bends the approximation so that it follows the surface over a much wider area (right).

An earlier Note on boosting ([XGBoost maths Note](#)) did this with one variable: it called the second derivative of one **observation's** (G-1374) loss (an observation is one record, a row of the data table) its Hessian h_i and stopped the Taylor series at the second-order term. This Note does the same with many variables, in this order:

1. how one curve bends: the second derivative (Section 2);
2. the flat approximation of a surface: the tangent plane (Section 3);
3. the curved approximation, which needs second partial derivatives (Section 4);
4. the Hessian, the matrix that holds those second partial derivatives and tells the shape of the surface (Section 5);
5. the full Taylor series and its uses in ML (Sections 6 and 7).

The Note uses the gradient of the [partial derivatives and gradients Note](#) and the eigenvalues of the [eigenvectors and eigenvalues Note](#).

2. The second derivative: how a curve bends

Key point: The second derivative is the rate of change of the slope: positive where a curve bends upward, negative where it bends downward, and larger where the bend is sharper.

The derivative of a curve is its slope. The slope is itself a function, so it has a derivative too: the **second derivative**, written $f''(x)$ or $\frac{d^2f}{dx^2}$. Where a curve bends upward its slope is increasing, so $f'' > 0$. Where it bends downward its slope is decreasing, so $f'' < 0$. A sharper bend means a larger f'' .

Figure 2 reads the second derivative off two steps. Take two equal steps dx to the right and compare the two rises.

1. The first step raises f by df_1 , the second by df_2 .
2. The difference $df_2 - df_1$ is the change in the change. On a straight line it is 0; on a curve that bends upward the second rise is bigger.
3. The second derivative is that difference divided by dx^2 , in the limit of tiny steps.

For $f(x) = x^3$ from $x = 1$ with $dx = 0.01$: $df_1 = 0.030301$ and $df_2 = 0.030907$. The difference is 0.000606, and $0.000606/0.01^2 = 6.06$, close to $f''(1) = 6x = 6$.

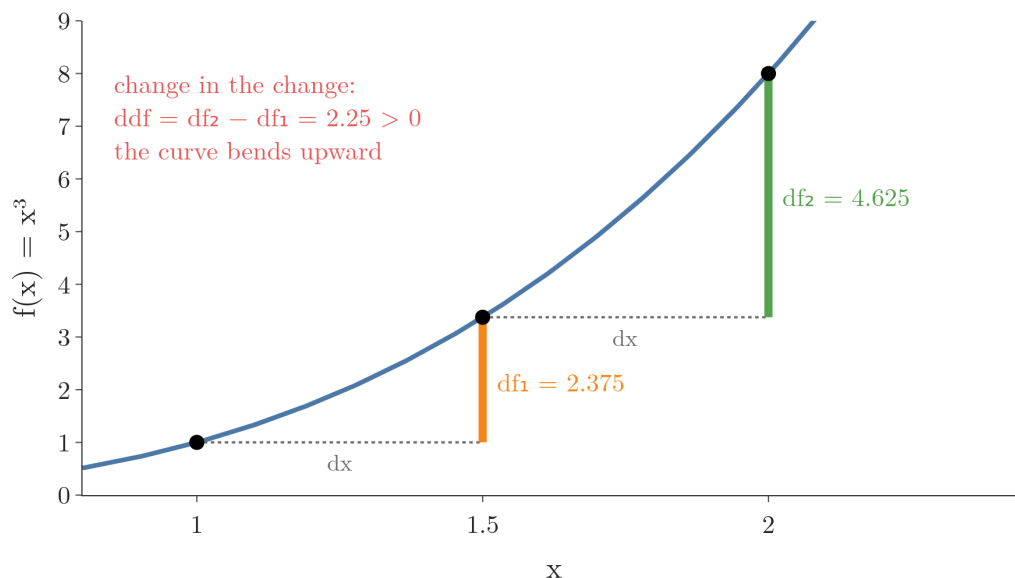


Figure 2: The second derivative as the change in the change, for $f(x) = x^3$ from $x = 1$. Two equal steps (drawn large, $dx = 0.5$) raise f by 2.375 and then by 4.625; the second rise is bigger, so the curve bends upward. Idea after 3Blue1Brown, “Higher order derivatives”

An everyday reading: if f is the distance a car has covered, f' is its velocity and f'' is its acceleration. A positive second derivative is the push back into the seat while the car speeds up; a negative one is braking.

The [derivatives of one variable Note](#) (Section 6.1) used f'' to make a parabola bend like a curve. The rest of this Note does the same for a surface.

3. Linearisation: the tangent plane

Key point: Near $\mathbf{x}_0$, $f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$: a flat plane that touches the surface at $\mathbf{x}_0$.

The [derivatives of one variable Note](#) replaced a curve near a point by its tangent line. With several inputs, the gradient gives a tangent plane (Figure 1, left): the plane spanned by the tangent lines of the two slices in the [partial derivatives and gradients Note](#).

1. **In words:** start from the value at $\mathbf{x}_0$ and add the gradient times the step. The gradient is a row and the step a column, so their product is one number.

2. **Formula:**

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla_{\mathbf{x}} f(\mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0)$$

3. **Example:** $f(x, y) = x^3 + xy + y^2$ at $(1, 1)$: $f(1, 1) = 3$ and $\nabla f(1, 1) = [3 + 1, 1 + 2] = [4, 3]$. At $(1.1, 0.9)$, a step of $(0.1, -0.1)$:

$$f(1.1, 0.9) \approx 3 + 4 \times 0.1 + 3 \times (-0.1) = 3.1 \quad (\text{true: } 3.131)$$

The step $\mathbf{x} - \mathbf{x}_0$ is what makes the plane touch the surface: at $\mathbf{x} = \mathbf{x}_0$ the step is zero, the second term vanishes, and the approximation equals $f(\mathbf{x}_0)$ exactly. The formula has the same form for 2 inputs or for 100.

The approximation is good close to $\mathbf{x}_0$ and gets worse further away, because a plane cannot bend. In Figure 1 (left), the orange plane touches the blue surface at the black dot and separates from it in every direction.

Figure 3 measures how far. It colours the size of the error at every point near $(1, 1)$. For the tangent plane (left), the error stays below 0.05 only inside a small oval around the point. The second-order polynomial of Section 4 (right) stays below 0.05 over a band about nine times as large. Its only missing term is δ_x^3 (Section 6.3), so its error depends on x alone, which is why the band has straight sides.

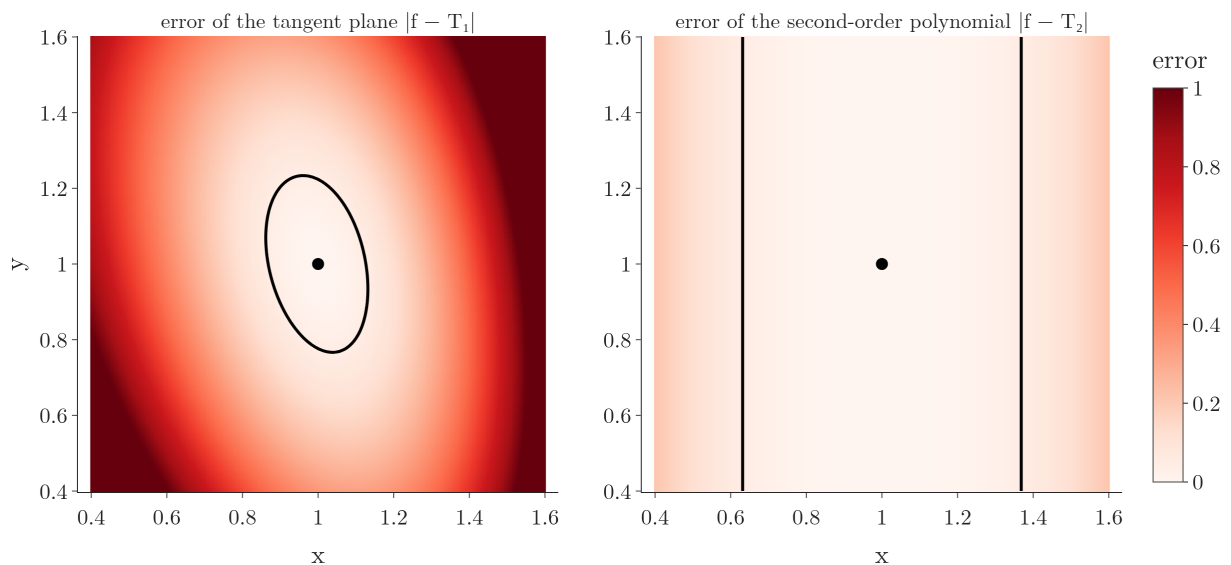


Figure 3: The size of the error near $(1, 1)$, on the same colour scale. Left: the tangent plane, $|f - T_1|$. Right: the second-order Taylor polynomial, $|f - T_2|$. The black line marks an error of 0.05; the region inside it is about nine times larger for T_2 .

4. Bending the plane: the quadratic approximation

Key point: Add the three terms δ_x^2 , $\delta_x\delta_y$ and δ_y^2 to the tangent plane, and choose their constants so that the second partial derivatives match those of f at the point.

A plane cannot bend, so the tangent plane leaves the surface quickly. The fix is a surface that can bend: the green surface of Figure 1 (right). It touches f at the point like the plane, and it also curves with f , so it stays close over a much wider area. Cut it with any vertical plane and the cut is a parabola.

Write the step as $\delta_x = x - x_0$ and $\delta_y = y - y_0$. The tangent plane uses the step once (δ_x, δ_y) . The simplest terms that bend are the ones that use it twice:

$$Q(x, y) = \underbrace{f(\mathbf{x}_0) + f_x \delta_x + f_y \delta_y}_{\text{tangent plane}} + a \delta_x^2 + b \delta_x \delta_y + c \delta_y^2$$

Here f_x and f_y are short for the two partial derivatives at the point. Such a Q is a **quadratic approximation** (G-2252). The three new terms are zero at the point, and so are their first partial derivatives, so they do not disturb what the tangent plane already matches: the value and the gradient. The constants a , b and c are free. To choose them we need second derivatives of a function of two variables.

4.1 Second partial derivatives

Key point: Differentiating a partial derivative again gives a second partial derivative; for smooth functions, the order of the two differentiations does not matter.

A partial derivative is itself a function of all the inputs, so we can differentiate it again. For a function $f(x, y)$ of two variables the notation is:

- $\partial^2 f / \partial x^2$: differentiate with respect to x twice;
- $\partial^2 f / \partial y \partial x$, the derivative with respect to y of $\partial f / \partial x$: first with respect to x , then with respect to y (read right to left);
- $\partial^n f / \partial x^n$: differentiate n times with respect to x .

Each of these is a **second partial derivative** (G-1760). The ones that mix two different variables are **mixed partial derivatives** (G-1236).

1. **In words:** for a function whose second derivatives are continuous, differentiating by x then y gives the same as by y then x .
2. **Formula:**

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

3. **Example:** $f(x, y) = x^3 + xy + y^2$, our running example. Its first partial derivatives are $\partial f / \partial x = 3x^2 + y$ and $\partial f / \partial y = x + 2y$. Then:

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 2, \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}(3x^2 + y) = 1, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}(x + 2y) = 1$$

The two mixed derivatives agree.

Figure 4 shows what “agree” means. On the left, we stand at $x = 1$ and watch the x -slope $\partial f / \partial x$ as y moves: it rises 1 for every unit of y . On the right, we stand at $y = 1$ and watch the y -slope as x moves: it also rises 1 per unit. How the x -slope responds to y equals how the y -slope responds to x .

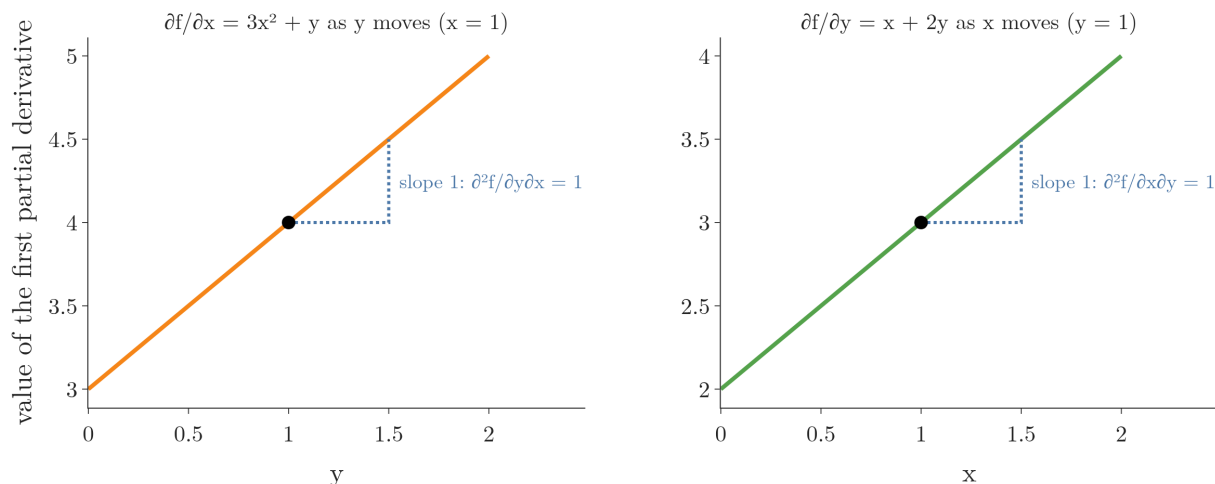


Figure 4: Mixed partial derivatives of $f(x, y) = x^3 + xy + y^2$ near $(1, 1)$. Left: the x-slope $3x^2 + y$ as y moves, with $x = 1$ fixed: slope 1. Right: the y-slope $x + 2y$ as x moves, with $y = 1$ fixed: also slope 1.

4.2 Fitting the three constants

Key point: Matching the second partial derivatives gives $a = \frac{1}{2}f_{xx}$, $b = f_{xy}$ and $c = \frac{1}{2}f_{yy}$.

We want Q to bend like f at the point, so we ask for the same three second partial derivatives. For the running example at $(1, 1)$: $f = 3$, $\nabla f = [4, 3]$, and from Section 4.1 the second partial derivatives are 6, 1 and 2. So

$$Q = 3 + 4\delta_x + 3\delta_y + a\delta_x^2 + b\delta_x\delta_y + c\delta_y^2$$

1. **Twice by x .** Only $a\delta_x^2$ survives two derivatives by x , and it gives $2a$. Matching $\partial^2 f / \partial x^2 = 6$ gives $a = 3$.
2. **By x , then by y .** Only $b\delta_x\delta_y$ survives, and it gives b . Matching the mixed partial derivative 1 gives $b = 1$.
3. **Twice by y .** Only $c\delta_y^2$ survives, and it gives $2c$. Matching $\partial^2 f / \partial y^2 = 2$ gives $c = 1$.

The result is $Q = 3 + 4\delta_x + 3\delta_y + 3\delta_x^2 + \delta_x\delta_y + \delta_y^2$. Check at $(1.1, 0.9)$, a step of $(0.1, -0.1)$: the plane gave 3.1, and the new terms add $3(0.01) + (-0.01) + 0.01 = 0.03$, so $Q = 3.13$ against the true 3.131.

In general, with f_{xx} , f_{xy} and f_{yy} short for the three second partial derivatives at the point,

$$a = \frac{1}{2}f_{xx}, \quad b = f_{xy}, \quad c = \frac{1}{2}f_{yy}$$

The halves cancel the 2 that the power rule brings down from the squares, exactly as for the parabola of one variable. The mixed term has no square, so it has no half.

Three numbers control the bend of a function of two inputs. With n inputs there are many more, and they need a tidy home: the Hessian.

5. The Hessian

Key point: The Hessian is the square, symmetric matrix of all second partial derivatives; it describes the **curvature** (G-521) of f near a point.

5.1 The Hessian matrix

Key point: Entry (i, j) is $\partial^2 f / \partial x_i \partial x_j$; for n inputs the Hessian is $n \times n$.

Figure 5 lays out every second partial derivative of the running example as a tree. Each arrow differentiates once, by x or by y . Two arrows in a row give four results, and the two mixed paths in the middle meet in the same value, as Section 4.1 showed. The four results, written in a grid, are the Hessian.

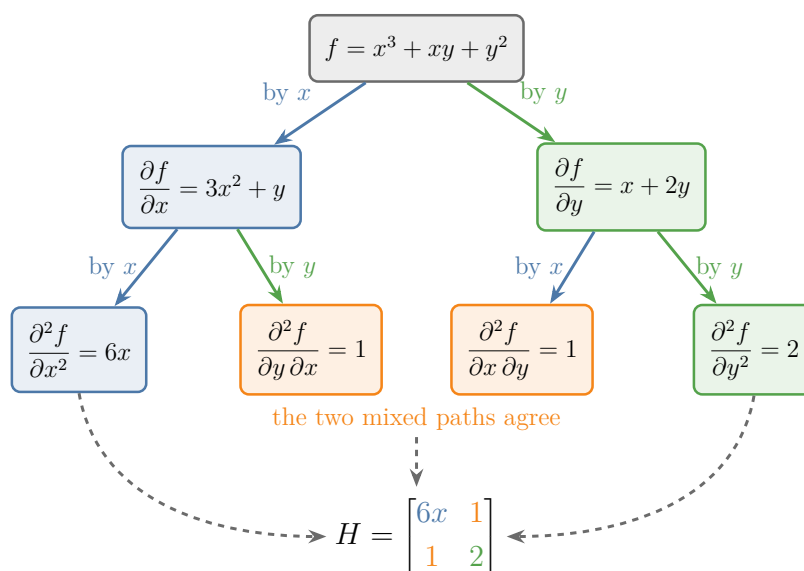


Figure 5: The tree of derivatives of $f = x^3 + xy + y^2$. Two differentiations give four second partial derivatives; the two mixed paths agree. The four results fill the Hessian. Idea after Khan Academy, “Symmetry of second partial derivatives” and “The Hessian matrix”; our own function

1. **In words:** the **Hessian matrix** (G-888) arranges all second partial derivatives in a grid, row i and column j holding the derivative with respect to x_i and x_j .
2. **Formula:** for two variables,

$$H = \nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ it is an $n \times n$ matrix, symmetric because the mixed derivatives are equal.

3. **Example:** for $f(x, y) = x^3 + xy + y^2$,

$$H = \begin{bmatrix} 6x & 1 \\ 1 & 2 \end{bmatrix}, \quad H(1, 1) = \begin{bmatrix} 6 & 1 \\ 1 & 2 \end{bmatrix}$$

The Hessian is a matrix-valued function: put in a point, get a matrix of numbers. It also writes the three bending terms of Section 4.2 as one product. With the step as a column $\delta = [\delta_x, \delta_y]^\top$,

$$\frac{1}{2} \delta^\top H \delta = \frac{1}{2} (6\delta_x^2 + 2\delta_x\delta_y + 2\delta_y^2) = 3\delta_x^2 + \delta_x\delta_y + \delta_y^2$$

the same $a = 3$, $b = 1$, $c = 1$. The mixed entry sits in the matrix twice, which is why b needed no half. Section 6.1 uses this form.

The Hessian is also the Jacobian of the gradient: the gradient is a function with n outputs, and differentiating each of them by each input gives $n \times n$ numbers (see the [Jacobian Note](#)). With a single input, the Hessian is just the second derivative, the h_i of the [XGBoost maths Note](#).

Extra: For a vector-valued function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, each of the m outputs has its own $n \times n$ Hessian. Stacked, they form an $m \times n \times n$ tensor.

5.2 Two Hessians we already know

Key point: The least-squares loss has the constant Hessian $2\Phi^\top\Phi$; the bowl of the gradient Note has the constant Hessian with rows $[2, 1]$ and $[1, 4]$.

Quadratic functions have a constant Hessian: the same at every point.

- **The bowl** $x_1^2 + x_1x_2 + 2x_2^2$ of the [partial derivatives and gradients Note](#) has gradient $[2x_1 + x_2, x_1 + 4x_2]$. Differentiating again:

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

- **The least-squares loss** $\|\mathbf{y} - \Phi\boldsymbol{\theta}\|^2$ has gradient $2\boldsymbol{\theta}^\top\Phi^\top\Phi - 2\mathbf{y}^\top\Phi$ (the [Jacobian Note](#), Section 7). Its Jacobian is

$$H = 2\Phi^\top\Phi$$

For the three-point data there (Φ with rows $[1, 1]$, $[1, 2]$, $[1, 3]$), $\Phi^\top\Phi$ has rows $[3, 6]$ and $[6, 14]$, so H has rows $[6, 12]$ and $[12, 28]$.

The matrix $X^\top X$ of the normal equation in the [multiple linear regression maths Note](#) is, up to the factor 2, the Hessian of the loss.

5.3 Curvature: what the Hessian says about the shape

Key point: The signs of the Hessian's eigenvalues give the shape near a flat point: all positive is a bowl (minimum), all negative a cap (maximum), mixed signs a saddle.

Why the shape matters. Training a model means finding the lowest point of a loss. At the lowest point the tangent plane is flat: the gradient is zero. A point with zero gradient is a **stationary point** (G-1879). A flat tangent plane is not enough, though. The top of a hill is flat too, and so is the middle of a horse's saddle. The second derivatives tell the three apart.

Slices. Cut the surface with a vertical plane through the point. The cut is a curve of one variable, and its second derivative says whether that slice bends up or down. Take $f = x^2 - y^2$ at the origin, where the gradient $[2x, -2y]$ is zero.

- The slice along x (keep $y = 0$) is x^2 : it bends up, so from this side the point looks like a minimum.
- The slice along y (keep $x = 0$) is $-y^2$: it bends down, so from this side the point looks like a maximum.

The two directions disagree. Such a point is a **saddle point** (G-1718): neither a minimum nor a maximum. A curve of one variable cannot do this, because it has only one direction. Figure 6 turns the slicing plane through every direction. Along a direction at angle θ to the x -axis the slice is $\cos(2\theta) s^2$, with s the distance along the slice, so its second derivative is $2 \cos 2\theta$: $+2$ along x , 0 on the diagonal, -2 along y .

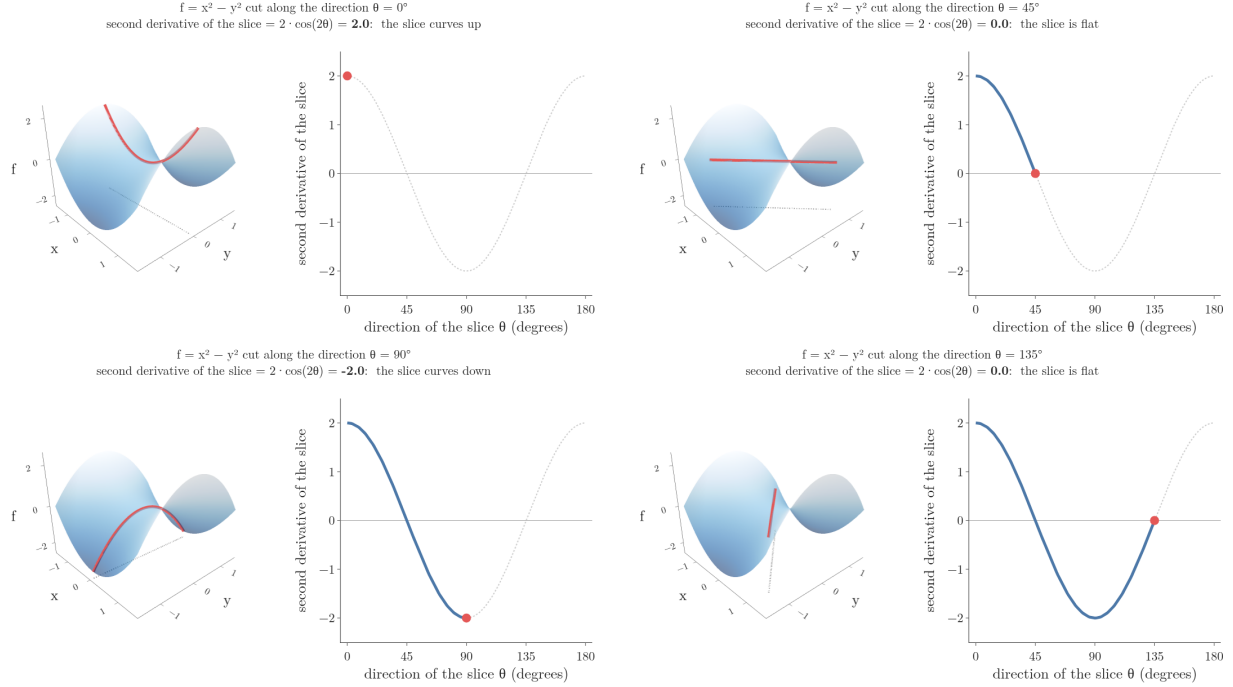


Figure 6: The saddle $f = x^2 - y^2$ cut by a vertical plane that turns from the x direction (angle 0) to the y direction (90 degrees). Left: the slice (red) on the surface. Right: the second derivative of the slice, $2 \cos 2\theta$, traced against the angle: positive, zero, then negative. Idea after Khan Academy, “Saddle points”

The two axis slices are not enough. Take $f = x^2 + y^2 + pxy$ with a number p we can turn up. Its second partial derivatives along the axes are $f_{xx} = 2$ and $f_{yy} = 2$ for every p : both axis slices always bend up. Yet Figure 7 shows the bowl turning into a saddle as p grows.

1. The slice along the diagonal direction $(1, -1)/\sqrt{2}$ is $\left(1 - \frac{p}{2}\right) s^2$, with second derivative $2 - p$.
2. For $p < 2$ it still bends up: a bowl.
3. At $p = 2$ it is flat.
4. For $p > 2$ it bends down while the axis slices bend up: a saddle.

The term that does this is the mixed one: $f_{xy} = p$. The mixed partial derivative measures how much the two diagonal directions disagree. The purest case is $f = xy$: flat along both axes ($f_{xx} = f_{yy} = 0$), bending up along one diagonal and down along the other.

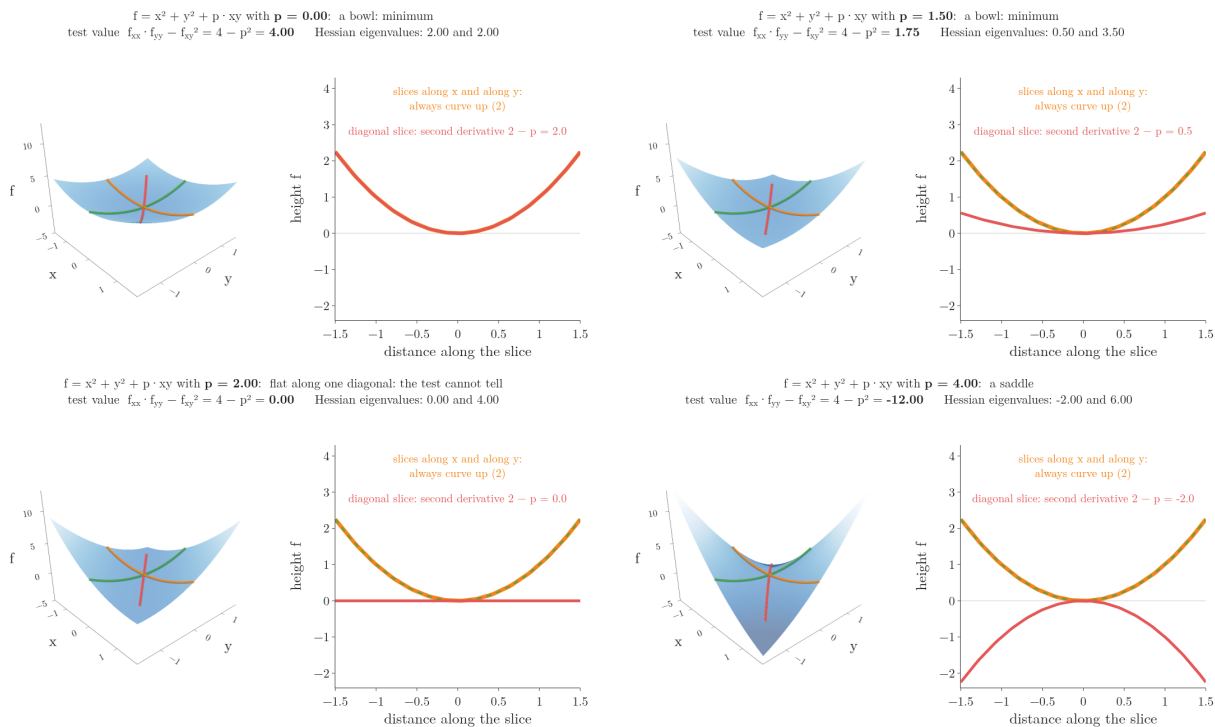


Figure 7: The surface $f = x^2 + y^2 + pxy$ as p grows from 0 to 4. The slices along x and y (orange, green) never change; the diagonal slice (red) flattens at $p = 2$ and then bends down, so the bowl becomes a saddle. The title tracks the test value $4 - p^2$ and the Hessian's eigenvalues $2 - p$ and $2 + p$. Idea after Khan Academy, “Second partial derivative test”

The second partial derivative test (G-2253). One number weighs the axis slices against the mixed term.

1. **In words:** multiply the two axis bends and subtract the square of the mixed partial derivative.
2. **Formula:** at a stationary point, compute

$$D = f_{xx} f_{yy} - f_{xy}^2$$

- $D > 0$ and $f_{xx} > 0$: a minimum;
 - $D > 0$ and $f_{xx} < 0$: a maximum;
 - $D < 0$: a saddle point;
 - $D = 0$: the test gives no answer.
3. **Example:** for $f = x^2 + y^2 + pxy$ at the origin, $D = 2 \cdot 2 - p^2 = 4 - p^2$. With $p = 0$, $D = 4$ and $f_{xx} = 2 > 0$: a minimum. With $p = 4$, $D = -12$: a saddle. The switch is at $p = 2$, where $D = 0$, as in Figure 7.

The running example $f = x^3 + xy + y^2$ has two stationary points. Setting $3x^2 + y = 0$ and $x + 2y = 0$ gives $(0, 0)$ and $(\frac{1}{6}, -\frac{1}{12})$. At $(0, 0)$ the Hessian has rows $[0, 1]$ and $[1, 2]$, so $D = 0 \cdot 2 - 1 = -1$: a saddle point. At $(\frac{1}{6}, -\frac{1}{12})$ it has rows $[1, 1]$ and $[1, 2]$, so $D = 2 - 1 = 1$ with $f_{xx} = 1 > 0$: a minimum.

The same test with eigenvalues. D is the determinant of the Hessian. For a 2×2 matrix the determinant is the product of the two eigenvalues, so $D > 0$ says the two eigenvalues have the same sign and $D < 0$ says their signs differ. The eigenvalue form works for any number of inputs, where a single number D is no longer enough. The Hessian's eigenvectors are the slice directions of strongest and weakest bending, and its eigenvalues are the second derivatives of those two slices (see the [eigenvectors and eigenvalues Note](#)). For $x^2 + y^2 + pxy$ the Hessian has rows $[2, p]$ and $[p, 2]$; its eigenvectors are the two diagonals and its eigenvalues are $2 + p$ and $2 - p$, the red slice of Figure 7. Figure 8 shows the three basic shapes.

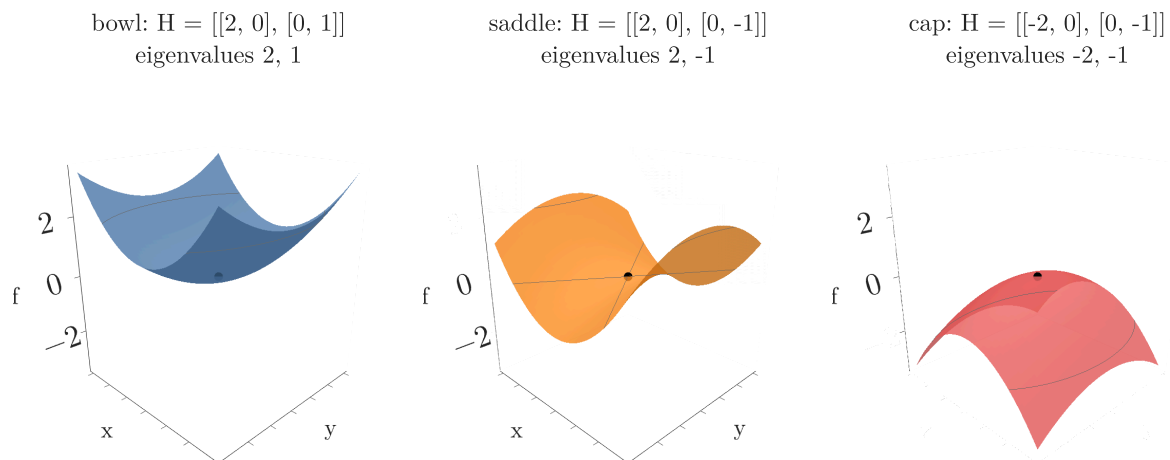


Figure 8: Three surfaces and the eigenvalues of their Hessians: both positive (bowl), opposite signs (saddle), both negative (cap)

- **Bowl:** all eigenvalues positive. The surface curves up in every direction; a point with zero gradient is a minimum.
- **Saddle** (a saddle point when the gradient is zero): positive and negative eigenvalues. The surface curves up in some directions and down in others; a point with zero gradient is neither a minimum nor a maximum.
- **Cap:** all eigenvalues negative. The surface curves down everywhere; a point with zero gradient is a maximum.

For the bowl of the gradient Note, the eigenvalues of H are $3 \pm \sqrt{2}$, that is 4.41 and 1.59. Both are positive, so the function is a bowl, as its contour map showed. The ratio $4.41/1.59 = 2.8$ says the bowl curves almost three times as steeply in one direction as in the other: its contour ellipses are stretched. The narrow valleys that slow down gradient descent in the [gradient descent Note](#) (Section 9) are Hessians with a large eigenvalue ratio.

Extra: A function whose Hessian has no negative eigenvalues at any point is convex (Boyd and Vandenberghe §3.1.4): a single bowl with no local minima to get stuck in, the property the [gradient descent Note](#) (Section 8) asked of a loss. The least-squares Hessian $2\Phi^T\Phi$ never has negative eigenvalues, because $\delta^T\Phi^T\Phi\delta = \|\Phi\delta\|^2 \geq 0$ for every δ . This Hessian is why linear regression's loss is convex.

6. The multivariate Taylor series

Key point: Add terms built from ever higher derivatives at $\mathbf{x}_0$, multiplied by ever more copies of the step $\boldsymbol{\delta} = \mathbf{x} - \mathbf{x}_0$; the second-order term is $\frac{1}{2}\boldsymbol{\delta}^\top H \boldsymbol{\delta}$.

6.1 The terms up to second order

Key point: Value, plus gradient times step, plus half the step times Hessian times step.

The tangent plane is the start of a series, exactly as in one variable, and the quadratic approximation of Section 4 is its next step. Section 4.2 found the bending terms one constant at a time; with the Hessian they are one line. With the step $\boldsymbol{\delta} = \mathbf{x} - \mathbf{x}_0$:

Order k	Term	Shapes	One-variable version
0	$f(\mathbf{x}_0)$	number	$f(x_0)$
1	$\nabla f(\mathbf{x}_0) \boldsymbol{\delta}$	$(1 \times D)(D \times 1)$	$f'(x_0) \delta$
2	$\frac{1}{2}\boldsymbol{\delta}^\top H(\mathbf{x}_0) \boldsymbol{\delta}$	$(1 \times D)(D \times D)(D \times 1)$	$\frac{1}{2}f''(x_0) \delta^2$

1. **In words:** the second-order term weighs every pair of step components $\delta_i \delta_j$ by the matching Hessian entry, adds them up and halves the sum.
2. **Formula:** the second-order **multivariate Taylor polynomial** (G-1284) is

$$T_2(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0) \boldsymbol{\delta} + \frac{1}{2} \boldsymbol{\delta}^\top H(\mathbf{x}_0) \boldsymbol{\delta}, \quad \boldsymbol{\delta}^\top H \boldsymbol{\delta} = \sum_i \sum_j H_{ij} \delta_i \delta_j$$

3. **Example:** at $(1, 1)$ with $\boldsymbol{\delta} = (0.1, -0.1)$ and H with rows $[6, 1]$ and $[1, 2]$:

$$\boldsymbol{\delta}^\top H \boldsymbol{\delta} = 6(0.1)^2 + 2 \cdot 1 \cdot (0.1)(-0.1) + 2(-0.1)^2 = 0.06 - 0.02 + 0.02 = 0.06$$

so $T_2 = 3.1 + \frac{1}{2} \times 0.06 = 3.13$, against the true 3.131.

Figure 1 (right) shows T_2 : a curved surface that follows f far beyond the region where the tangent plane was good.

Python: The three terms in NumPy.

```
import numpy as np

f0 = 3.0
g = np.array([4.0, 3.0])           # gradient at (1, 1)
H = np.array([[6.0, 1.0], [1.0, 2.0]]) # Hessian at (1, 1)
d = np.array([0.1, -0.1])         # step to (1.1, 0.9)

f0 + g @ d                        # 3.1   (first order)
f0 + g @ d + 0.5 * d @ H @ d    # 3.13  (second order)
```

6.2 Higher orders: tensors and outer products

Key point: The k -th term uses a k -index array of k -th derivatives and k copies of the step; for $k = 2$ these are the Hessian and the outer product $\delta\delta^\top$.

The pattern continues. The k -th derivative at $\mathbf{x}_0$, written $D^k f(\mathbf{x}_0)$, has one entry for every choice of k input indices: a $D \times \cdots \times D$ tensor with k indices. The tensor is combined with k copies of δ .

1. **In words:** multiply each k -th derivative entry by the matching product of step components, add them all, and divide by $k!$.
2. **Formula:** the **multivariate Taylor series** (G-1285) is

$$f(\mathbf{x}) = \sum_{k=0}^{\infty} \frac{D^k f(\mathbf{x}_0) \delta^k}{k!}, \quad D^k f(\mathbf{x}_0) \delta^k = \sum_{i_1=1}^D \cdots \sum_{i_k=1}^D D^k f(\mathbf{x}_0)[i_1, \dots, i_k] \delta_{i_1} \cdots \delta_{i_k}$$

Stopping after the $k = n$ term gives the Taylor polynomial T_n .

3. **Example:** for $k = 2$, δ^2 means the **outer product** (G-1418) $\delta\delta^\top$, the matrix with entries $\delta_i \delta_j$. With $\delta = (0.1, -0.1)$:

$$\delta\delta^\top = \begin{bmatrix} 0.01 & -0.01 \\ -0.01 & 0.01 \end{bmatrix}$$

Multiplying entry by entry with H and adding: $6(0.01) + 1(-0.01) + 1(-0.01) + 2(0.01) = 0.06$, the same $\delta^\top H \delta$ as before.

Each extra copy of δ adds one index: $\delta\delta^\top$ is a matrix, and $\delta \otimes \delta \otimes \delta$, with entries $\delta_i \delta_j \delta_k$, is a 3-index tensor (see the [tensors Note](#)).

6.3 A complete example: the expansion is exact for a polynomial

Key point: A polynomial of degree 3 needs exactly the terms up to third order; then the Taylor polynomial is the function itself.

As in one variable (the [derivatives of one variable Note](#), Section 6.4), the Taylor series of a polynomial stops. For $f(x, y) = x^3 + xy + y^2$ at $(1, 1)$, write $\delta_x = x - 1$ and $\delta_y = y - 1$.

- **Order 0:** $f(1, 1) = 3$.
- **Order 1:** $\nabla f(1, 1) = [4, 3]$ gives $4\delta_x + 3\delta_y$.
- **Order 2:** $\frac{1}{2}\delta^\top H \delta = \frac{1}{2}(6\delta_x^2 + 2\delta_x \delta_y + 2\delta_y^2) = 3\delta_x^2 + \delta_x \delta_y + \delta_y^2$.
- **Order 3:** of all third partial derivatives only $\partial^3 f / \partial x^3 = 6$ is not zero, so the term is $6\delta_x^3 / 3! = \delta_x^3$.
- **Order 4 and up:** all derivatives are zero.

1. **In words:** add the four orders.

2. **Formula:**

$$f(x, y) = 3 + 4\delta_x + 3\delta_y + 3\delta_x^2 + \delta_x \delta_y + \delta_y^2 + \delta_x^3$$

3. **Example:** at $(1.1, 0.9)$ the running totals are $T_1 = 3.1$, $T_2 = 3.13$, $T_3 = 3.13 + 0.1^3 = 3.131$, now exactly $f(1.1, 0.9) = 1.331 + 0.99 + 0.81 = 3.131$. Further away, at $(1.5, 0.5)$, they are 3.5, 4.25 and $4.375 = f(1.5, 0.5)$: the low orders are worse there, but the full expansion is still exact.

Multiplying out the brackets gives back $x^3 + xy + y^2$; we checked this symbolically.

Figure 9 walks along the straight path from $(1, 1)$ to $(1.5, 0.5)$ and reads all three polynomials on the way. Near the start they agree with f . Further out the tangent plane T_1 falls behind first, then T_2 ; T_3 stays on f the whole way.

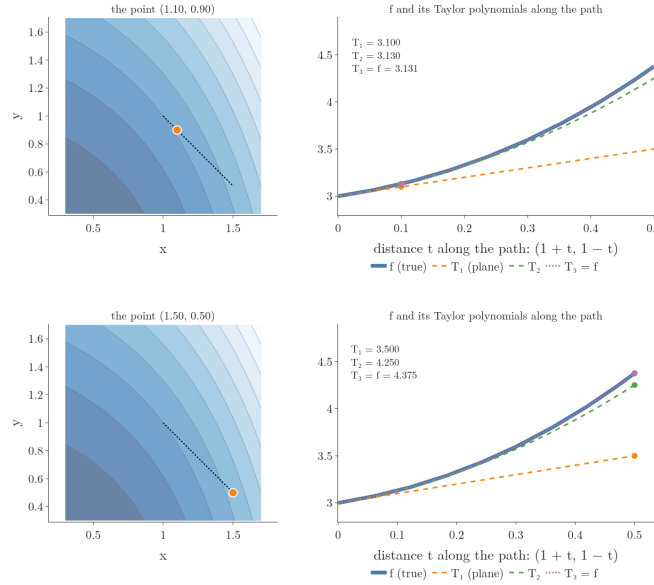


Figure 9: The Taylor polynomials of $f(x, y) = x^3 + xy + y^2$ around $(1, 1)$ along the path to $(1.5, 0.5)$. Left: the moving point on the contour map. Right: f (thick blue) against T_1 , T_2 and T_3 ; T_3 equals f .

7. Where ML uses second-order approximations

Key point: Minimising the second-order Taylor polynomial instead of the function itself is **Newton's method** (G-1321); XGBoost does this for every tree, and other methods use it to approximate distributions.

The second-order polynomial is a bowl (when H has positive eigenvalues), and the bottom of a bowl has a formula. The formula makes T_2 a useful stand-in for a function that is hard to minimise directly.

Extra: Newton's method in several variables. Setting the gradient of T_2 with respect to δ to zero gives $\nabla f^\top + H\delta = \mathbf{0}$, so the step to the bottom of the local bowl is

$$\delta = -H^{-1} \nabla f^\top$$

For the bowl of the gradient Note at $(1, 1)$, $\nabla f = [3, 5]$ and H has rows $[2, 1]$ and $[1, 4]$. Solving $H\boldsymbol{\delta} = -[3, 5]^\top$ gives $\boldsymbol{\delta} = (-1, -1)$, landing exactly on the minimum $(0, 0)$ in one step: for a quadratic function T_2 is the function itself. Gradient descent, which only uses the tangent plane, needs many small steps for the same trip. With one variable this is the **Newton step** of the [gradient boosting classification Note](#), and XGBoost’s leaf formula $-G/(H + \lambda)$ in the [XGBoost maths Note](#) is a Newton step per leaf (Chen and Guestrin §2.2). The catch: with millions of parameters the Hessian has millions squared entries, which is why deep learning mostly stays with first-order methods (Goodfellow et al. §8.6.1).

Extra: Quasi-Newton methods (G-1603): **BFGS** (G-283) and **L-BFGS** (G-1023). These skip the Hessian and build a stand-in matrix B from gradients alone. After each step $\mathbf{s} = \mathbf{x}_{k+1} - \mathbf{x}_k$, the gradient change $\mathbf{y} = \nabla f_{k+1} - \nabla f_k$ is measured, and B is updated so that

$$B_{k+1} \mathbf{s} = \mathbf{y}$$

The condition is the **secant equation** (G-1757): the new B must reproduce the gradient change just seen. With one variable it says $B = (f'(x_{k+1}) - f'(x_k))/(x_{k+1} - x_k)$, the slope between two gradient readings. For $f = x^3$ stepping from $x = 1$ to $x = 2$: f' goes from 3 to 12, so $B = (12 - 3)/(2 - 1) = 9$, between the true curvatures $f''(1) = 6$ and $f''(2) = 12$. The step is then $\boldsymbol{\delta} = -B^{-1}\nabla f^\top$, as in Newton’s method, usually shortened by a line search. **BFGS** (Broyden, Fletcher, Goldfarb, Shanno) is the most used update rule; it keeps B symmetric and positive definite, so every step goes downhill. **L-BFGS** (“limited memory”) stores only the last few $(\mathbf{s}, \mathbf{y})$ pairs instead of the full $n \times n$ matrix, which makes it usable with many parameters. L-BFGS is the default solver of **LogisticRegression** in the [logistic regression hyperparameters Note](#). (Nocedal and Wright, ch. 6 for BFGS, §7.2 for L-BFGS.)

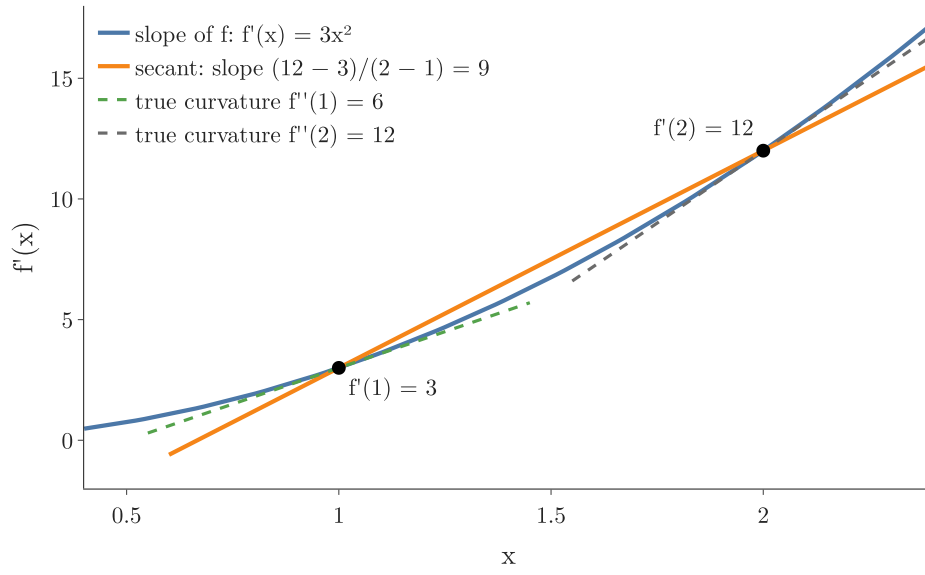


Figure 10: The secant idea for $f = x^3$: the line through the slope readings at $x = 1$ and $x = 2$ has slope 9, a stand-in for the true curvatures 6 and 12

Figure 10 shows the one-variable case: BFGS never computes the curvature (dashed), it reads the slope twice and takes the line through the two readings (orange).

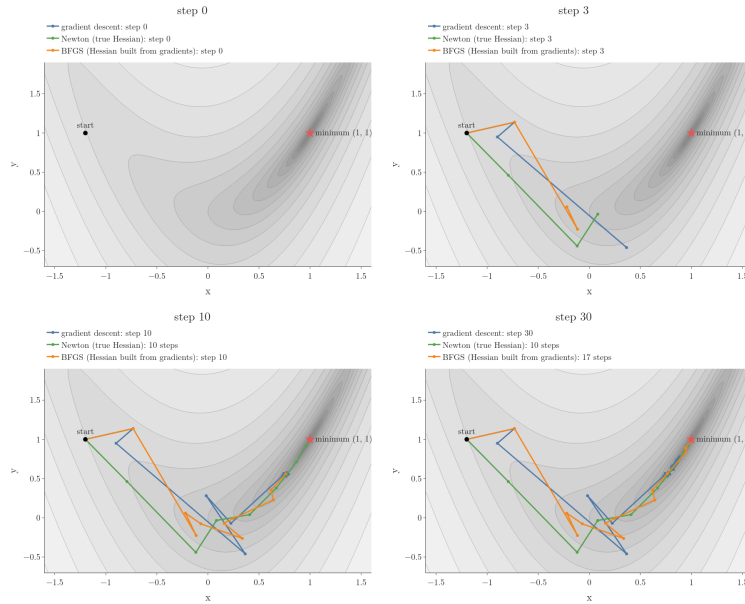


Figure 11: Gradient descent, Newton and BFGS walking down the same curved valley from $(-1.2, 1)$ to the minimum $(1, 1)$, one step per frame

Figure 11 races the three methods on the curved valley $f = (1 - x)^2 + 5(y - x^2)^2$, all with the same rule for the step length. Newton, with the true Hessian, needs 10 steps. BFGS,

which only ever sees gradients, needs 17: its first steps wander while B is still a guess, then it settles into the valley like Newton. The figure script checks this: for the first 7 steps B is still 90% or more away from the true Hessian, and from step 7 on it is within about 30%. Gradient descent zig-zags across the narrow valley and needs 717 steps.

Extra: Second-order Taylor expansions also approximate probability distributions. The **Laplace approximation** (G-1043) replaces a distribution near its peak by a normal distribution whose spread comes from the Hessian of its log there (Bishop §4.4). The **extended Kalman filter**, used to track moving objects, linearises a nonlinear system at every time step with the first-order expansion (Thrun et al. §3.3).

8. Summary

Object	Formula	Example: $f = x^3 + xy + y^2$ at $(1, 1)$
Mixed partial derivatives	$\partial^2 f / \partial x \partial y = \partial^2 f / \partial y \partial x$	both equal 1
Quadratic approximation	tangent plane $+ \frac{1}{2} f_{xx} \delta_x^2 + f_{xy} \delta_x \delta_y + \frac{1}{2} f_{yy} \delta_y^2$	3.13 at $(1.1, 0.9)$
Hessian	$H_{ij} = \partial^2 f / \partial x_i \partial x_j$, symmetric $n \times n$	rows $[6, 1]$, $[1, 2]$
Second partial derivative test	$D = f_{xx} f_{yy} - f_{xy}^2$ at a stationary point	$(0, 0)$: $D = -1$, saddle; $(\frac{1}{6}, -\frac{1}{12})$: $D = 1$, minimum
Tangent plane (first order)	$f(\mathbf{x}_0) + \nabla f \boldsymbol{\delta}$	3.1 at $(1.1, 0.9)$
Second-order Taylor	$+ \frac{1}{2} \boldsymbol{\delta}^T H \boldsymbol{\delta}$	3.13
Third-order Taylor	$+ \frac{1}{3!} D^3 f \boldsymbol{\delta}^3$	3.131 (exact)
Newton step	$\boldsymbol{\delta} = -H^{-1} \nabla f^T$	bowl: $(1, 1) \rightarrow (0, 0)$
Secant equation (BFGS)	$B_{k+1} \mathbf{s} = \mathbf{y}$	x^3 , $1 \rightarrow 2$: $B = 9$

- Second partial derivatives can be taken in either order; the Hessian is therefore symmetric.
- The Hessian measures curvature; the signs of its eigenvalues separate bowls, saddles and caps. For two inputs the second partial derivative test reads the same from $f_{xx} f_{yy} - f_{xy}^2$.
- The mixed partial derivative can turn a bowl into a saddle even when both axis slices bend up.
- Linearisation replaces a surface by its tangent plane, good only near the point.
- The multivariate Taylor series adds terms with k copies of the step and the k -th derivative tensor; the second-order term is $\frac{1}{2} \boldsymbol{\delta}^T H \boldsymbol{\delta}$.
- A polynomial of degree k is reproduced exactly by its Taylor polynomial of degree k .

9. Sources

Built from

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Chapter 5.
- 3Blue1Brown, “Higher order derivatives | Chapter 10, Essence of calculus”, YouTube, <https://www.youtube.com/watch?v=BLkz5LGWihw> (Figure 2)
- Khan Academy, “What is a tangent plane”, YouTube, https://www.youtube.com/watch?v=cHNT7_F8
- Khan Academy, “Local linearization”, YouTube, https://www.youtube.com/watch?v=o7_zS7Bx2VA
- Khan Academy, “What do quadratic approximations look like”, YouTube, https://www.youtube.com/watch?v=80bJA_tSbo4
- Khan Academy, “Quadratic approximation formula, part 1”, YouTube, https://www.youtube.com/watch?v=80bJA_tSbo4
- Khan Academy, “Quadratic approximation formula, part 2”, YouTube, https://www.youtube.com/watch?v=80bJA_tSbo4
- Khan Academy, “Symmetry of second partial derivatives”, YouTube, <https://www.youtube.com/watch?v=L2buigM> (Figure 5)

- Khan Academy, “The Hessian matrix”, YouTube, <https://www.youtube.com/watch?v=LbBcuZukCAw> (Figure 5)
- Khan Academy, “Vector form of multivariable quadratic approximation”, YouTube, <https://www.youtube.com/watch?v=ClFrIg0PpnM>
- Khan Academy, “Multivariable maxima and minima”, YouTube, <https://www.youtube.com/watch?v=u>
- Khan Academy, “Saddle points”, YouTube, https://www.youtube.com/watch?v=8aAU4r_pUUU (Figure 6)
- Khan Academy, “Second partial derivative test”, YouTube, <https://www.youtube.com/watch?v=m1Fh> (Figure 7)
- Khan Academy, “Second partial derivative test intuition”, YouTube, <https://www.youtube.com/watch?>

Other references

- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. Section 4.4, the Laplace approximation.
- Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Section 3.1.4, second-order conditions.
- Chen, T. and Guestrin, C. (2016). “XGBoost: A Scalable Tree Boosting System”. *KDD*. Section 2.2.
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Section 8.6.1, Newton’s method.
- Nocedal, J. and Wright, S. J. (2006). *Numerical Optimization*, 2nd ed. Springer. Chapter 6 and section 7.2.
- Thrun, S., Burgard, W. and Fox, D. (2005). *Probabilistic Robotics*. MIT Press. Section 3.3, the extended Kalman filter.

10. Key terms

Term	Meaning
Second derivative	The derivative of the derivative: the rate at which the slope changes; positive where a curve bends upward
Quadratic approximation (G-2252)	The tangent plane plus the terms $\frac{1}{2}f_{xx}\delta_x^2 + f_{xy}\delta_x\delta_y + \frac{1}{2}f_{yy}\delta_y^2$; the second-order Taylor polynomial
Second partial derivative	A partial derivative of a partial derivative, such as $\partial^2 f / \partial x^2$
Mixed partial derivative	A second partial derivative with respect to two different variables, such as $\partial^2 f / \partial y \partial x$
Hessian matrix	The symmetric $n \times n$ matrix of all second partial derivatives of $f : \mathbb{R}^n \rightarrow \mathbb{R}$; it measures curvature
Curvature	How fast the slope of a surface changes; given in each direction by the Hessian

Term	Meaning
Saddle point	A flat point where the surface curves up in some directions and down in others (Hessian eigenvalues of both signs)
Stationary point	A point where every partial derivative is zero: a minimum, a maximum or a saddle point
Second partial derivative test (G-2253)	At a stationary point of a function of two inputs, $D = f_{xx}f_{yy} - f_{xy}^2$: $D > 0$ a minimum or maximum (by the sign of f_{xx}), $D < 0$ a saddle point, $D = 0$ no answer
Tangent plane	The flat plane that touches a surface at a point with the same gradient; the first-order Taylor approximation
Multivariate Taylor series	$\sum_k D^k f(\mathbf{x}_0) \boldsymbol{\delta}^k / k!$: approximation of f near $\mathbf{x}_0$ from its derivatives there
Multivariate Taylor polynomial	The multivariate Taylor series cut after the $k = n$ term
Outer product	$\boldsymbol{\delta}\boldsymbol{\delta}^\top$, the matrix of all products $\delta_i\delta_j$; more copies give tensors
Newton's method	Repeatedly jumping to the minimum of the second-order Taylor polynomial: $\boldsymbol{\delta} = -H^{-1}\nabla f^\top$
Quasi-Newton method	Newton's method with the Hessian replaced by a matrix built from gradient changes
Secant equation	$B_{k+1}\mathbf{s} = \mathbf{y}$: the Hessian stand-in must reproduce the last gradient change
BFGS	The most used quasi-Newton update; keeps the Hessian stand-in symmetric and positive definite
L-BFGS	Limited-memory BFGS: keeps only the last few step and gradient-change pairs; scikit-learn's default logistic regression solver
Laplace approximation	Approximating a distribution near its peak by a normal distribution built from the Hessian